Technical Writing: LaTeX Demonstration

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1 Introduction

LaTeX is a document preparation system used for structured technical and scientific writing.

This document demonstrates **bold text**, *italic text*, and coloured text.

1.1 Learning Objectives

- Organise a document using sections.
- Typeset mathematical expressions.
- Create tables and figures.
- Use automatic cross-references.

2 Mathematical Expressions

Inline mathematics fits within a sentence: $a^2 + b^2 = c^2$.

A numbered equation appears separately:

$$\sum_{i=1}^{n} i = \frac{n(n+1)}{2} \tag{1}$$

Equation (1) gives the sum of the first n positive integers.

A matrix:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

A piecewise function:

$$f(x) = \begin{cases} -x, & x < 0, \\ x, & x \geq 0. \end{cases}$$

3 Tables

Table 1 compares three tools.

Table 1: Technical Writing Tools

Tool	Purpose
LaTeX	Preparing technical documents
Beamer	Creating presentations
Doxygen	Generating source-code documentation

4 Figures

Figure 1 is drawn using the picture environment, without an external image.



Figure 1: LaTeX compilation workflow

5 Compilation Procedure

1. Write and save the LaTeX source.
2. Compile using pdfLaTeX.
3. Check the output for errors.
4. Compile again if references need updating.

Key point: The source file contains commands and content. The compiled PDF displays the formatted document.

6 Conclusion

LaTeX provides consistent formatting and automatic numbering for structured documents. The LaTeX Project provides further information [1].

References

[1] The LaTeX Project. `https://www.latex-project.org/`